MOHAMED EYAD

Robotics Software Engineer (ROS 2, C++, Python) · Perception, Manipulation & Navigation
Genova, Italy (open to relocation) · +39 351 823 4947 · mooeyad@gmail.com
LinkedIn · GitHub · Portfolio

PROFILE

Robotics engineer building ROS 2 systems on real hardware: robot arms with MoveIt and 3D vision, LiDAR-based SLAM and navigation, and real-time sensor pipelines on NVIDIA Jetson. MSc Robotics Engineering at the University of Genova, with a master's thesis at ETH Zurich (SCAI Lab) and the DART Lab: a real-time ROS 2 system with on-device LLM feedback, validated in a 12-participant study. BSc Mechatronics, Ain Shams University (GPA 3.53, ranked 6th in class). Studied and worked in Egypt, the UK, Italy and Switzerland.

RESEARCH & WORK EXPERIENCE

Master's Thesis Researcher | SCAI Lab (ETH Zurich) & DART Lab – Lake Lucerne Institute, Vitznau, Switzerland *03/2026 – 08/2026*

- Built a wearable, real-time cognitive-load estimator on ROS 2, using multimodal sensor fusion of heart-rate/HRV (Polar H10) and pupil data (Pupil Labs Neon) into a live cognitive-load index during treadmill walking.
- Implemented closed-loop spoken feedback with a local LLM on a Jetson AGX Orin, adapting to the participant's live physiological state.
- Ran a 12-participant study: the index tracked the mental task in all 12 participants and agreed with NASA-TLX ratings (r = 0.80).
- Developed the Pupil Labs Neon driver and eye-tracking message types for the lab's open-source ROS 2 framework, ROS4Healthcare.

Researcher & Lab Engineer | Human Centered Mechatronics (HCM) Lab, Ain Shams University, Cairo *01/2024 – 07/2024*

- Ran biomechanical and gait studies with Qualisys motion capture, AMTI force plates and Delsys Trigno EMG; tested EEG systems for real-time brain-activity acquisition.
- Operated and maintained the Fourier Intelligence ExoMotus X2 exoskeleton with spinal-cord-injury patients, controlling its walking behaviour through ROS (admittance control) and troubleshooting faults.
- Reverse-engineered custom medical components with an EinScan HX 3D scanner and prototyped on a Raise3D E2 printer; supported graduation-project and internship teams.

Teaching Assistant (part-time) | Faculty of Engineering, Ain Shams University, Cairo *09/2023 – 08/2024*

- Teaching assistant for the "Design of Mechatronics" course (ROS, CoppeliaSim, Arduino, RTOS, Proteus, MATLAB/Simulink, SolidWorks, Inventor); supervised bachelor robotics graduation projects.
- Trainer of the summer 2024 ROS 1 course: ROS fundamentals, Raspberry Pi, OpenCV, rosserial and robot navigation in Gazebo.

Teaching Assistant | UPEI Cairo Campus *09/2023 – 01/2024*

- Prepared and delivered tutorials and lab sessions for "Control System Design" and "Project-Based Professional Practice I".

STEAM Facilitator & Robotics Instructor | Edu-Zone STEAM Learning Centre, Cairo *03/2023 – 10/2023*

- Designed robotics content and taught engineering concepts to children aged 4–14.

EDUCATION

MSc Robotics Engineering | University of Genova, Italy *09/2024 – 12/2026 (expected)*

BSc Mechatronics, Robotics & Automation Engineering | Faculty of Engineering, Ain Shams University, Cairo *2018 – 2023*

- Cumulative GPA 3.53 – ranked 6th in class. Graduation project: human–robot collaborative assembly with an ABB IRB 120 (A+).

Exchange Semester – Mechatronics & Intelligent Machines | UCLan, United Kingdom *01/2020 – 05/2020*

- Fully funded Erasmus+ exchange programme. Grade: Distinction.

TECHNICAL SKILLS

Robotics: ROS 1 / ROS 2 (Jazzy), Nav2, MoveIt, PlanSys2 (PDDL), SLAM, visual servoing, Gazebo (Harmonic), MuJoCo, CoppeliaSim, RViz, Docker, GitHub Actions CI, rostest, ABB EGM/RWS, OctoMap
Programming: Python (NumPy, Pandas, Matplotlib, Tkinter), C / C++, Linux (Ubuntu), Git, MATLAB / Simulink, LabVIEW
Control: PID and feedback control on real hardware, force/admittance/impedance control, sensor fusion and synchronisation
Perception & ML: OpenCV, PyTorch, YOLO, ArUco, COLMAP, 3D Gaussian Splatting, diffusion augmentation, local LLMs (Ollama)
Hardware & sensing: Jetson AGX Orin/Nano, STM32, ESP8266, Arduino/AVR, RealSense D435i, 2D LiDAR, IMUs, Polar H10, Pupil Labs Neon, PLC basics
Biomechanics lab: Qualisys motion capture, AMTI force plates, Delsys Trigno EMG, EEG, ExoMotus X2 exoskeleton
Design & electronics: SolidWorks, Autodesk Inventor, Altium Designer, Proteus, 3D printing & scanning
Languages: Arabic (native), English (fluent, C1), Italian (A2)

SELECTED PROJECTS

Collaborative Assembly Using a Serial Manipulator (ABB IRB 120) | Graduation project, team of 6 – grade A+ *07/2023*

- ABB IRB 120 controlled from ROS through EGM/RWS, with MoveIt planning and a custom pneumatic end effector; FreeCAD scripts read the Inventor CAD files to build the parts list.
- Two Intel RealSense D435i cameras: YOLOv7 on the wrist camera detects gears, bearings, bolts, nuts, flanges and springs (92.6% mAP@0.5); YOLOv8 on the environment camera segments people and obstacles and locates the hand for handovers.
- OctoMap-based re-planning around people and obstacles; Pocketsphinx voice commands and a Tkinter GUI for CAD analysis, camera views and robot control.

Tools: ROS, MoveIt, ABB EGM/RWS, OctoMap, Python, YOLOv7/v8, Intel RealSense D435i · GitHub · Video

Few-shot 3D Gaussian Splatting with Diffusion Augmentation | Computer vision / 3D reconstruction *08/2026*

- Few-shot novel-view synthesis using inpainting, outpainting and pose-guided diffusion augmentation, evaluated across 3 subset sizes and 4 synthetic-data ratios.

Tools: Python, COLMAP, 3D Gaussian Splatting, diffusion models · GitHub

Autonomous Inspection Robot with ROS 2, PlanSys2 & Nav2 | University of Genova *01/2026*

- PDDL task planning (PlanSys2) with C++ action executors driving Nav2 (SLAM Toolbox, AMCL) for a mobile robot that explores waypoints to detect, sort and inspect ArUco markers in Gazebo Harmonic.

Tools: ROS 2 Jazzy, PlanSys2, Nav2, OpenCV, Gazebo Harmonic · GitHub

ArUco Visual Servoing for a Mobile Robot | RICE Lab, University of Genova *12/2025*

- Finite-state machine with proportional control for ArUco detection, ID sorting and target centring, taken from Gazebo to a real Husarion ROSbot 2; published as an open-source ROS 2 package.

Tools: ROS 2 Jazzy, OpenCV, Gazebo Harmonic · GitHub

Cognitive Architecture for a Search-and-Rescue TIAGo | Team of 3, University of Genova *04/2025*

- Modular architecture for mapping, victim triage, structural-risk assessment and reporting, prototyped in ROS Noetic with UML design and custom messages. Main contributor: diagrams, RiskReport message, launch files, rosbag test data and rostest integration tests in GitHub Actions CI.

Tools: ROS Noetic, UML, rostest, GitHub Actions · GitHub

Step Scope – Wearable Gait Monitoring System | Team project – grade A *05/2023*

- Wireless IMU (MPU6050) + FSR system monitoring plantar pressure and gait for diabetic-foot and neurological patients: gait-event detection, shank kinematics, stability assessment, ESP8266 (NodeMCU) transmission and Tkinter GUI.

Tools: ROS, Python, Arduino IDE · GitHub · Project page

Rehabilitation Robotics – Control Testing Rig | Team project – grade A *01/2023*

- Single-joint DC-motor rig with encoder, load cell and current sensor; modelled in MATLAB/Simulink, then position, current, force, admittance and impedance control implemented on Arduino and compared on hardware.

Tools: Inventor, Arduino IDE, MATLAB · Project page

Robusta – Autonomous Service Robot for Coffee Shops | Team project – grade A *05/2022*

- Full design and build: FastSLAM (Gmapping) and ROS Navigation Stack with 2D LiDAR on Jetson Nano; EKF fusing a Bosch BNO055 IMU with wheel-encoder odometry; PID motor control on STM32 in C++ (Arduino framework); customer emotion recognition with computer vision.

Tools: ROS, Gazebo, RViz, SolidWorks, Inventor · GitHub

TRAINING & CERTIFICATIONS

- Internet of Things (IoT) and Artificial Intelligence (AI) – National Telecommunication Institute (NTI), Aug 2023.
- IoT Fundamentals: Connecting Things – Cisco Networking Academy, Sep 2023.
- ASmarine Academy (02/2022 – 07/2022): Python, GitHub, Arduino interfacing, MATLAB, introduction to ROS & Gazebo; final project: TurtleBot3 navigation in Gazebo.
- Embedded Systems Diploma (07/2021 – 01/2022): C / Embedded C, AVR microcontroller drivers.
- PLC & Industrial Automation Summer Training – Schneider Electric Egypt, 2020.

REFERENCES

Available upon request (master's thesis supervisor, SCAI Lab, ETH Zurich; graduation-project and teaching supervisors, Ain Shams University).